Homework — 2.3.1 Algorithms

Name: ____ Date: _ ***Mark:*** ___ / 25

OCR H446 — set after the 2.3.1 lesson. Show traces/working.

Part A — This week's topic (~14 marks)

1. A search algorithm halves the portion of a sorted list it is examining on every step.

(a) State its Big O time complexity. **[1]** *(AO1)*

(b) **Justify** this complexity in one sentence, and state the **precondition** the list must satisfy. **[1]** *(AO2)*

2. Trace a **bubble sort** (one pass = compare adjacent pairs left-to-right, swapping if out of order) on the list below. Show the list **after each complete pass** until it is sorted. **[3]** *(AO2)*

Start: `[6, 2, 9, 4, 1]`

After pass	Item 1	Item 2	Item 3	Item 4	Item 5
Pass 1					
Pass 2					
Pass 3					
Pass 4					

3. A **binary search** is performed on the sorted list below to find the value **31**.

`[4, 9, 15, 18, 23, 27, 31, 42, 56]` (indices 0–8)

Complete the trace table (use integer division for mid). Show each step until the value is found. **[3]** *(AO2)*

Step	low	high	mid (index)	value at mid	< = > target?
1					
2					

State the result: ______

4. Consider the binary search tree below.

```
          55
         /  \
       32    78
      /  \   /  \
    20   45 60   90
```

Write the output of each traversal. **[3]** *(AO2)*

(a) **In-order:** ______ **[1]**

(b) **Pre-order:** ______ **[1]**

(c) **Post-order:** ______ **[1]**

5. Dijkstra's algorithm is run on the weighted graph below, starting at node **A**, to find the shortest distance to every other node.

Edges (undirected, with weights): A–B = 2, A–C = 5, B–C = 1, B–D = 7, C–D = 3, C–E = 8, D–E = 2

Complete the table of **final shortest distances** from A. **[3]** *(AO2)*

Node	Shortest distance from A	Via (previous node)
A	0	—
B		
C		
D		
E		

Part B — Retrieval: keep it fresh (~6 marks)

6. *(2.3.1)* State whether **merge sort** or **quick sort** guarantees O(n log n) in the worst case, and give the **worst-case** complexity of the other. **[2]**

7. *(2.2.1)* State the data structure used by a **breadth-first traversal**, and explain why that structure is appropriate. **[2]**

8. *(2.2.2)* A problem can only be solved in **exponential time or worse**. State the term used to describe such a problem, and give **one** example. **[2]**

Part C — Exam-style question (~5 marks)

9. A logistics company needs to calculate the shortest delivery distance from a single depot to **every** delivery point on a road network where all distances are positive.

Recommend either **Dijkstra's algorithm** or the **A*** algorithm for this task, and **justify** your choice by comparing the two. **[5]** *(AO3)*